

(الإجابة النموذجية التفصيلية)

$$\begin{aligned} v_{Ai} &= 0 & v_{Bi} &= 0 \\ m_A &= m & m_B &= 2m \end{aligned}$$

2

1

$$\sum F_A = \sum F_B$$

$$\Delta t_A = \Delta t_B$$

$$\sum F_A \cdot \Delta t_A = \sum F_B \cdot \Delta t$$

$$I_A = I_B$$

$$\Delta p_A = \Delta p_B$$

$$p_{Af} - p_{Ai} = p_{Bf} - p_{Bi}$$

$$p_A = p_B$$

دفع A
قذيفة B

$$m_B = 30 \text{ kg}$$

$$v_{Bi} = 0$$

$$v_{Bf} = 100 \text{ m/s} \rightarrow X^+$$

3

2

$$\Delta p_A = -\Delta p_B = -m_B(v_{Bf} - v_{Bi})$$

$$= -30(100 - 0) = -3000 \text{ N.s}$$

X⁺

$$m_A = 2m_B$$

$$p_A = p_B$$

$$\frac{KE_A}{KE_B} = \frac{\frac{p_A^2}{2m_A}}{\frac{p_B^2}{2m_B}} = \frac{1}{2}$$

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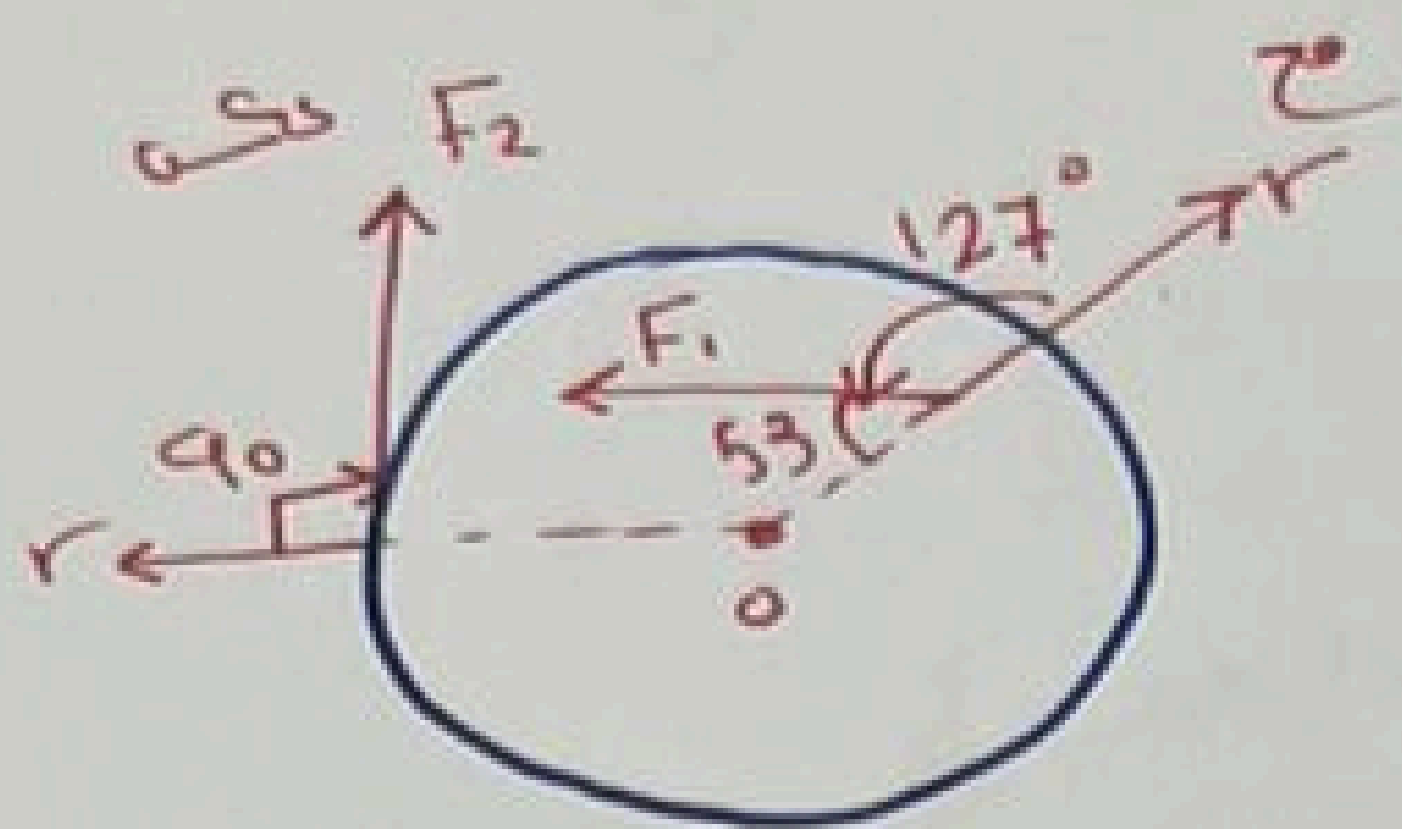
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4

$$\frac{KE_A}{KE_B} \neq \frac{1}{2} \rightarrow \frac{2}{2} KE_A = \frac{KE_B}{2}$$

1



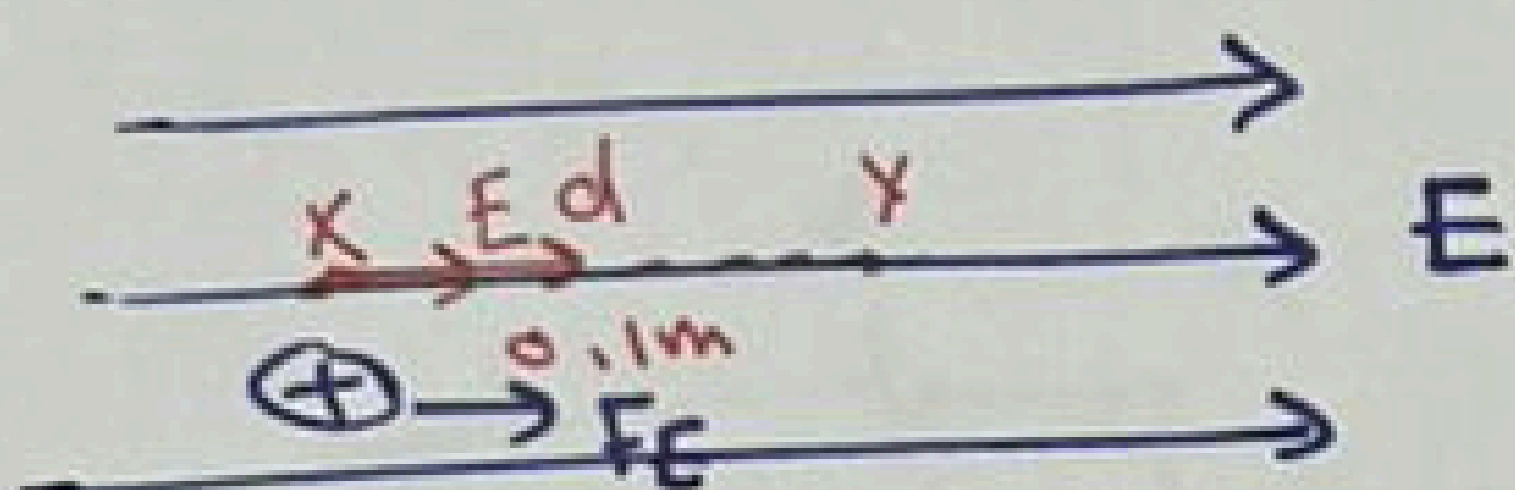
الحجم متوحد

$$V_{\text{مركز}} = V_{\text{مح}}$$

$$F_1 r_1 \sin 127 = F_2 r_2 \sin 90$$

$$15 \times 0.25 \times 0.8 = F_2 \times 0.3 \times 1$$

$$F_2 = 10 \text{ N}$$



$$\Delta PE = -8 \times 10^{-18} \text{ J}$$

بما أن المجال سالب يعني شغل قوة كهر بائية

$$W_{\text{كهربائية}} = -\Delta PE$$

$$W_{\text{كهربائية}} = -q(V_y - V_x)$$

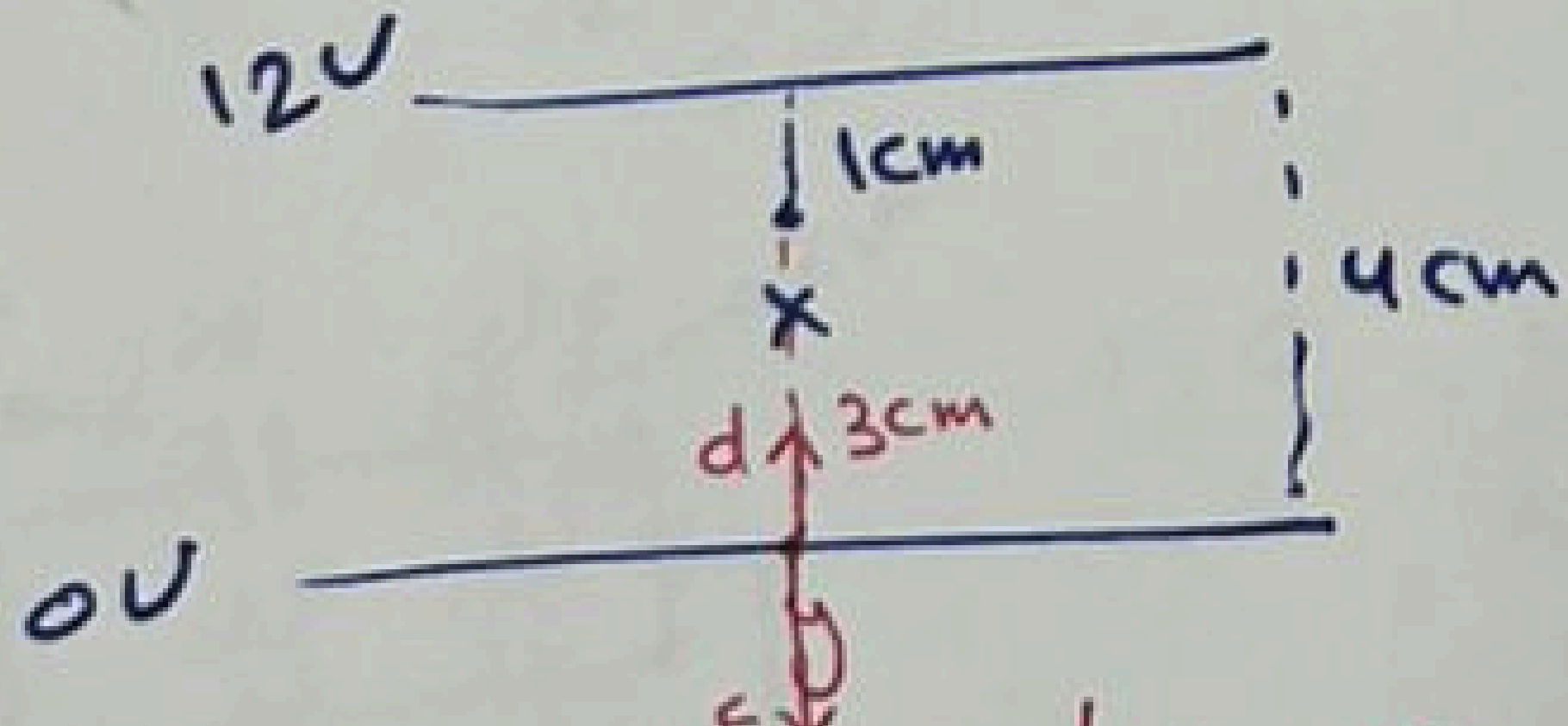
$$-8 \times 10^{-18} = -1.6 \times 10^{-19} (V_y - V_x)$$

$$V_y - V_x = 50 \text{ V}$$

$$V_y - V_x = -Ed \cos 0$$

$$-50 = -E(0.1)(1)$$

$$E = 500 \frac{\text{V}}{\text{m}}$$



$$V_x - V_y = -Ed \cos 180$$

$$V_x - 0 = 300(3 \times 10^{-2})(-1)$$

$$V_x - 0 = 9$$

$$V_x = 9 \text{ V}$$

$$E = \left| \frac{\Delta V}{d} \right| = \frac{12-0}{4 \times 10^{-2}} = 300 \frac{\text{V}}{\text{m}}$$

$$2$$

$$\gamma = FL$$

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ذراع المواء وبالرسم يسيرا الى اليمين d

$$C = \frac{\epsilon \cdot A}{d}$$

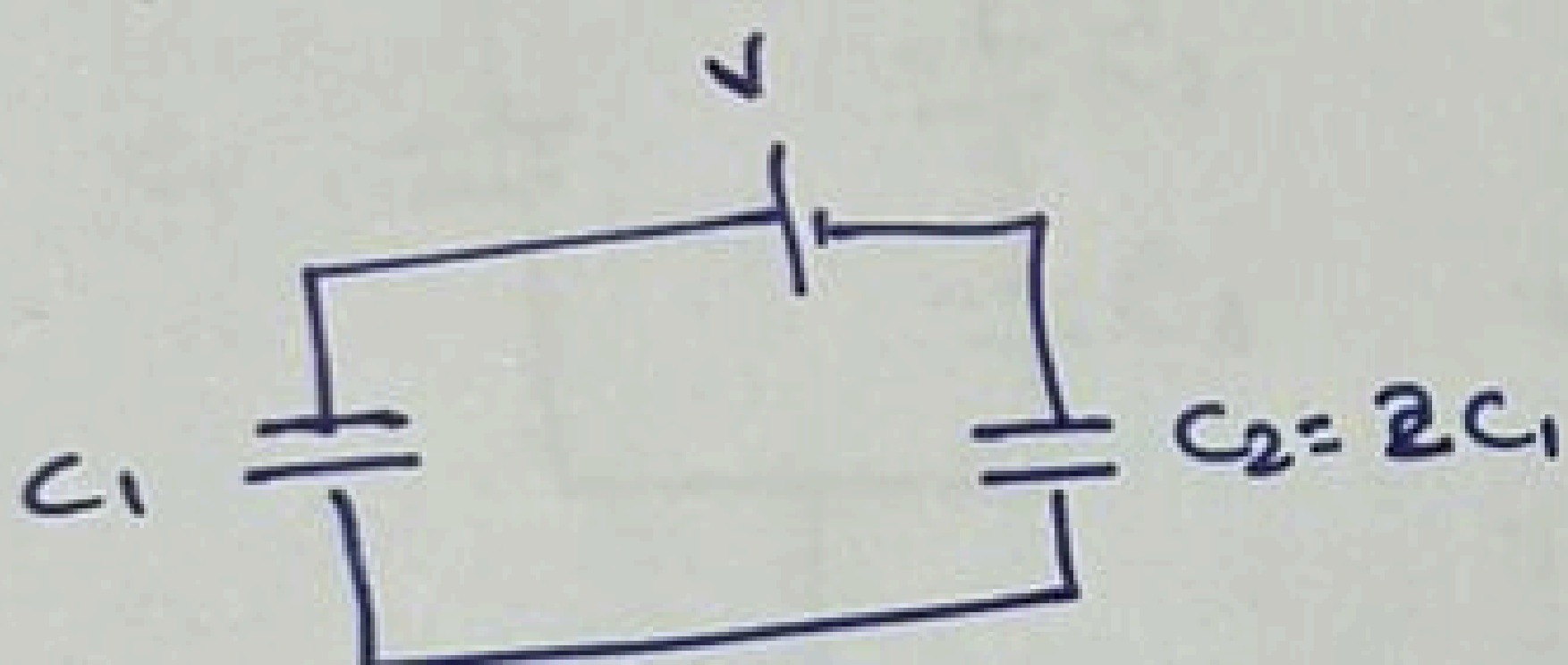
10

$$C_1 = \frac{\epsilon \cdot A}{d} = C$$

$$C_2 = \frac{\epsilon \cdot A}{2d} = \frac{C}{2}$$

$$C_3 = \frac{\epsilon \cdot 2A}{d} = 2C$$

$$C_4 = \frac{\epsilon \cdot 2A}{2d} = C$$



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$$V = V_1 + V_2 \quad (C_1, C_2 \text{ متوالي})$$

$$Q_1 = Q_2$$

$$V = \frac{Q}{C} \rightarrow V \propto \frac{1}{C}$$

$$V_1 = 2V_2$$

$$C_1 = C$$

$$C_2 = 2C$$

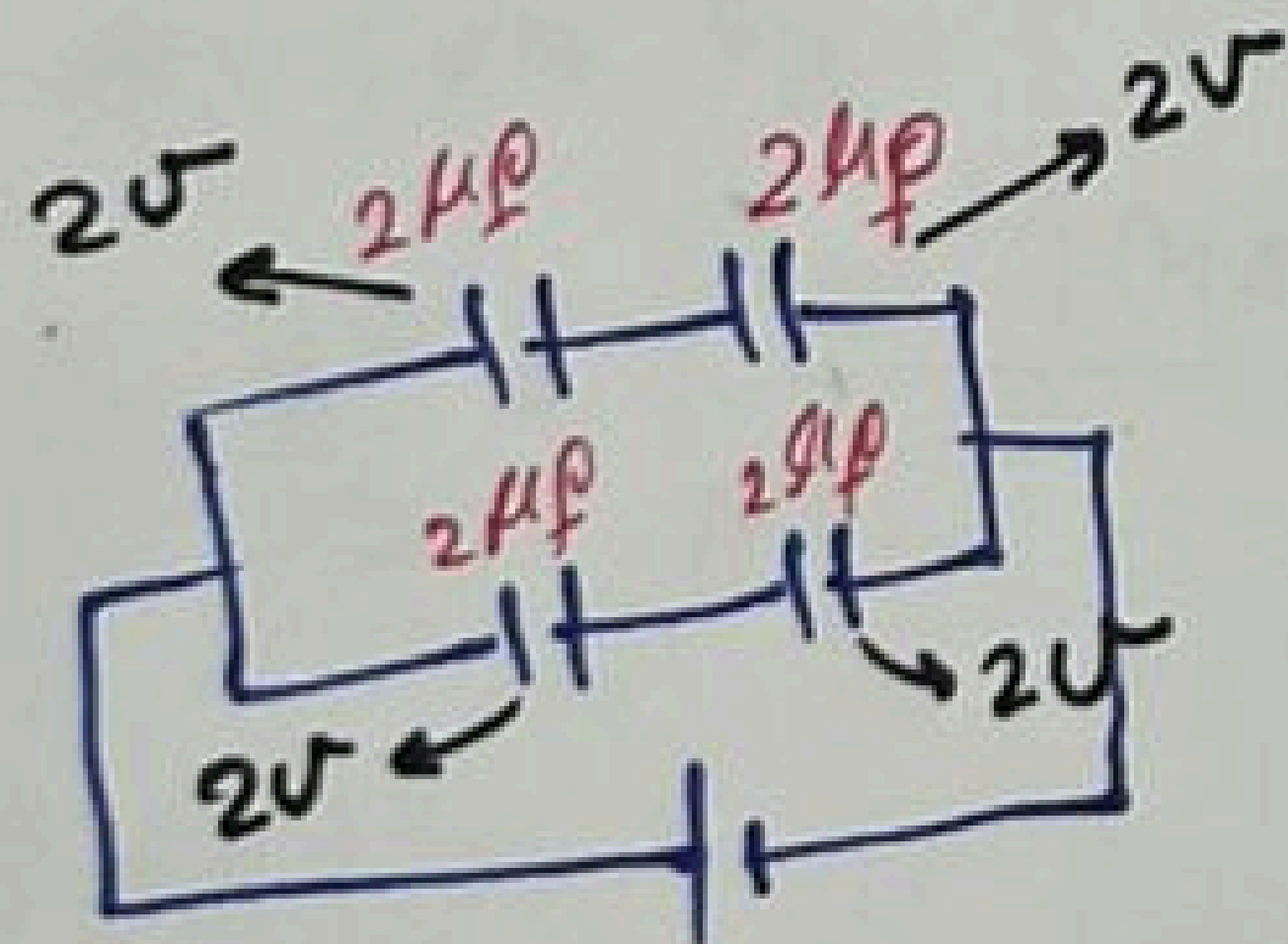
$$V_1 = V_2 \quad (\text{متوازي})$$

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$$\frac{PE_1}{PE_2} = \frac{\frac{1}{2} C_1 V^2}{\frac{1}{2} C_2 V^2} = \frac{1}{2}$$

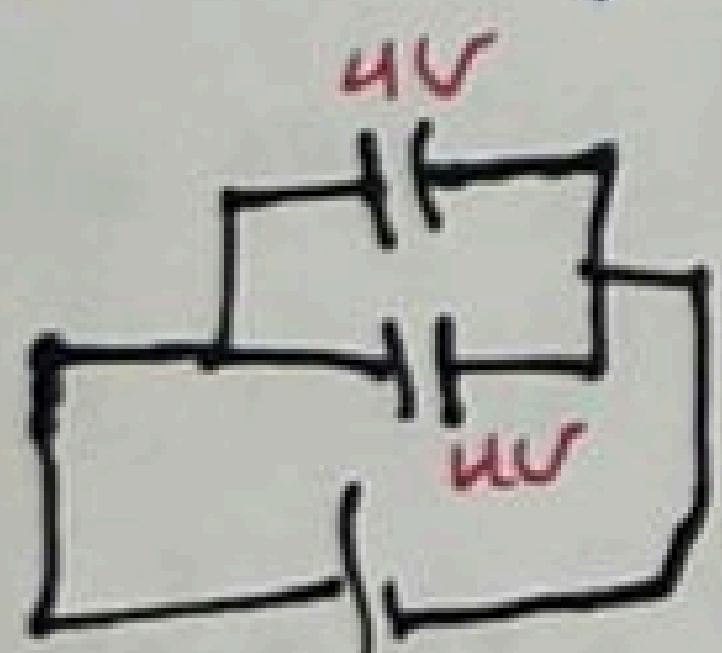
$$\rightarrow \frac{PE_1}{PE_2} \times \frac{1}{2} = PE_2 = 2PE_1$$

$$PE_2 = 2(9) = 18 \mu J$$



$$V = \frac{Q}{C} = \frac{4 \times 10^{-6}}{2 \times 10^{-6}} = 2V$$

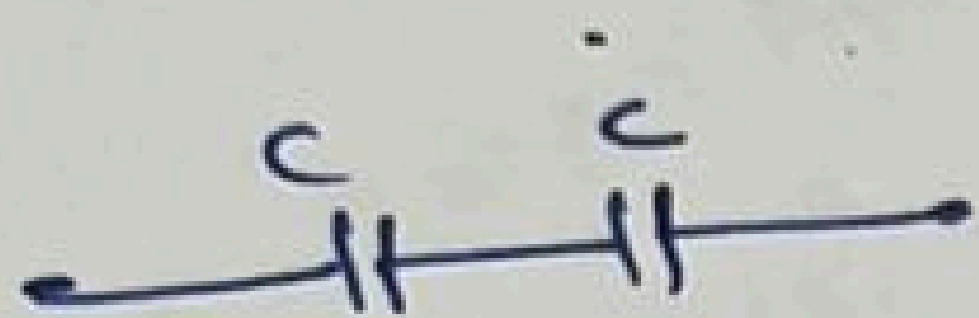
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$$V = V_1 = V_2 = V_3 = V_4 = 4V$$

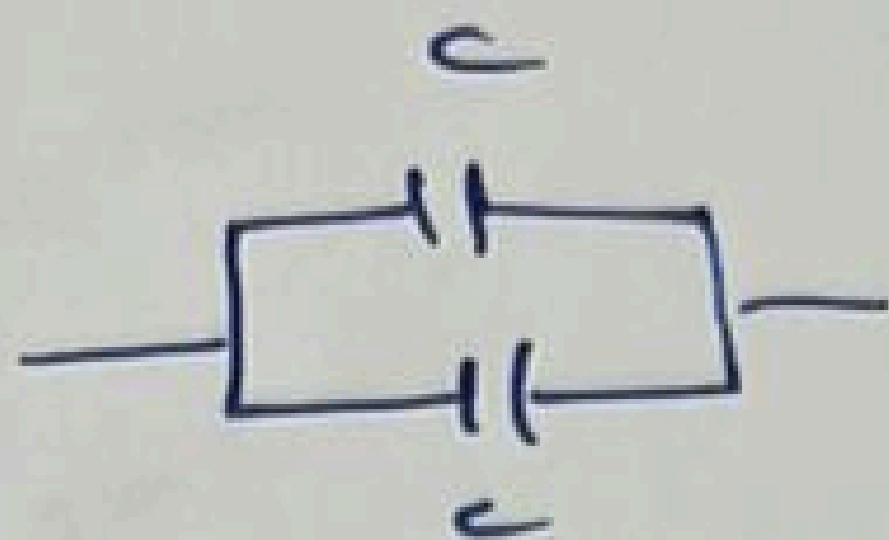
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3



$$C_{\text{eff}} = \frac{C}{2}$$

توازی

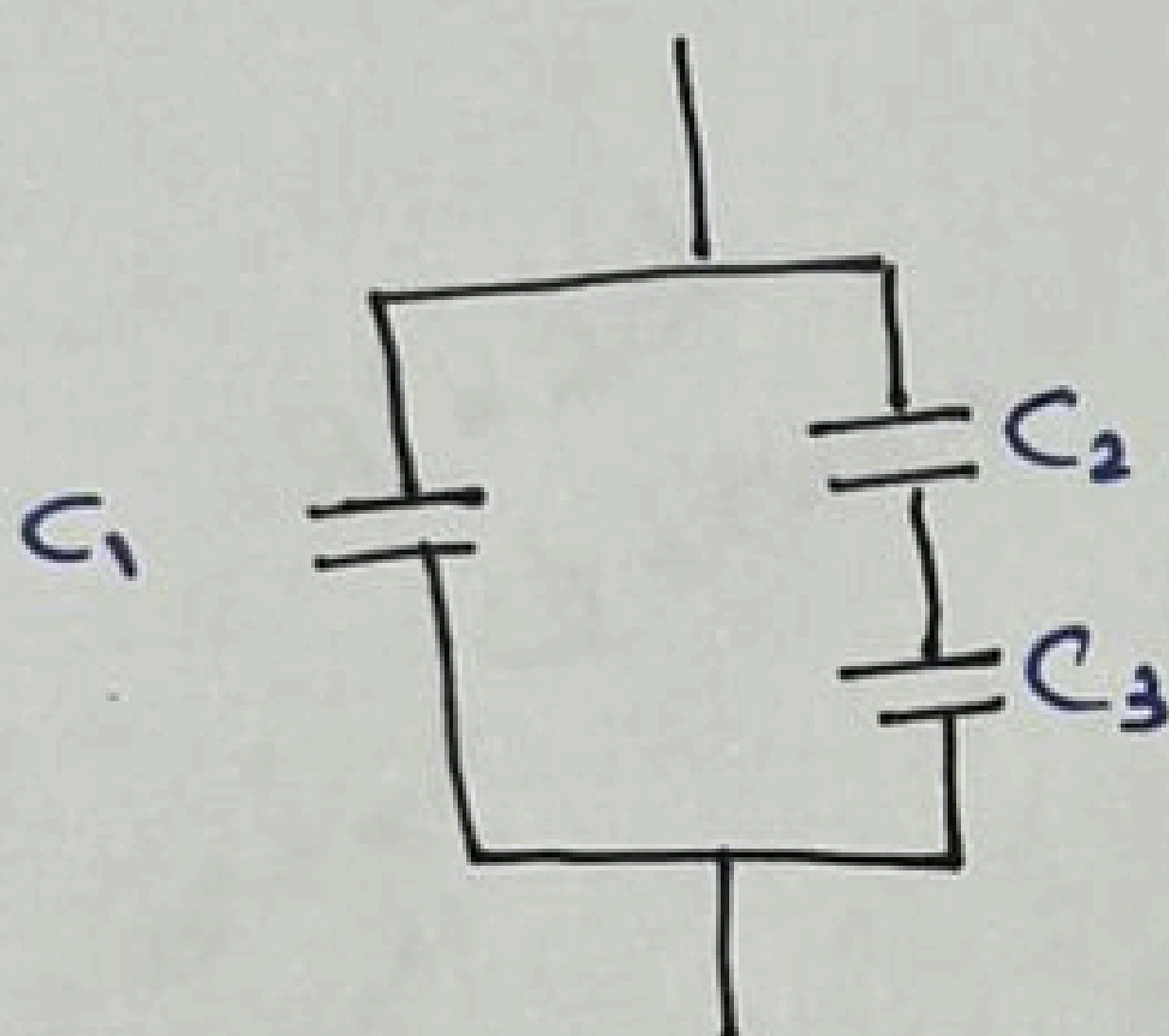


$$C_{\text{eff}} = 2C$$

توازی

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$$\frac{C_{\text{توازی}}}{C_{\text{توازی}}} = \frac{2C}{\frac{C}{2}} = \frac{4}{1}$$



$$Q_1 = 40 \text{ nC}$$

$$Q_2 = ?!$$

$$V_1 = V_2 + V_3$$

$$\frac{Q_1}{C} = \frac{Q_2}{C} + \frac{Q_3}{C}$$

$$Q_1 = Q_2 + Q_3$$

همان الکتریسیته

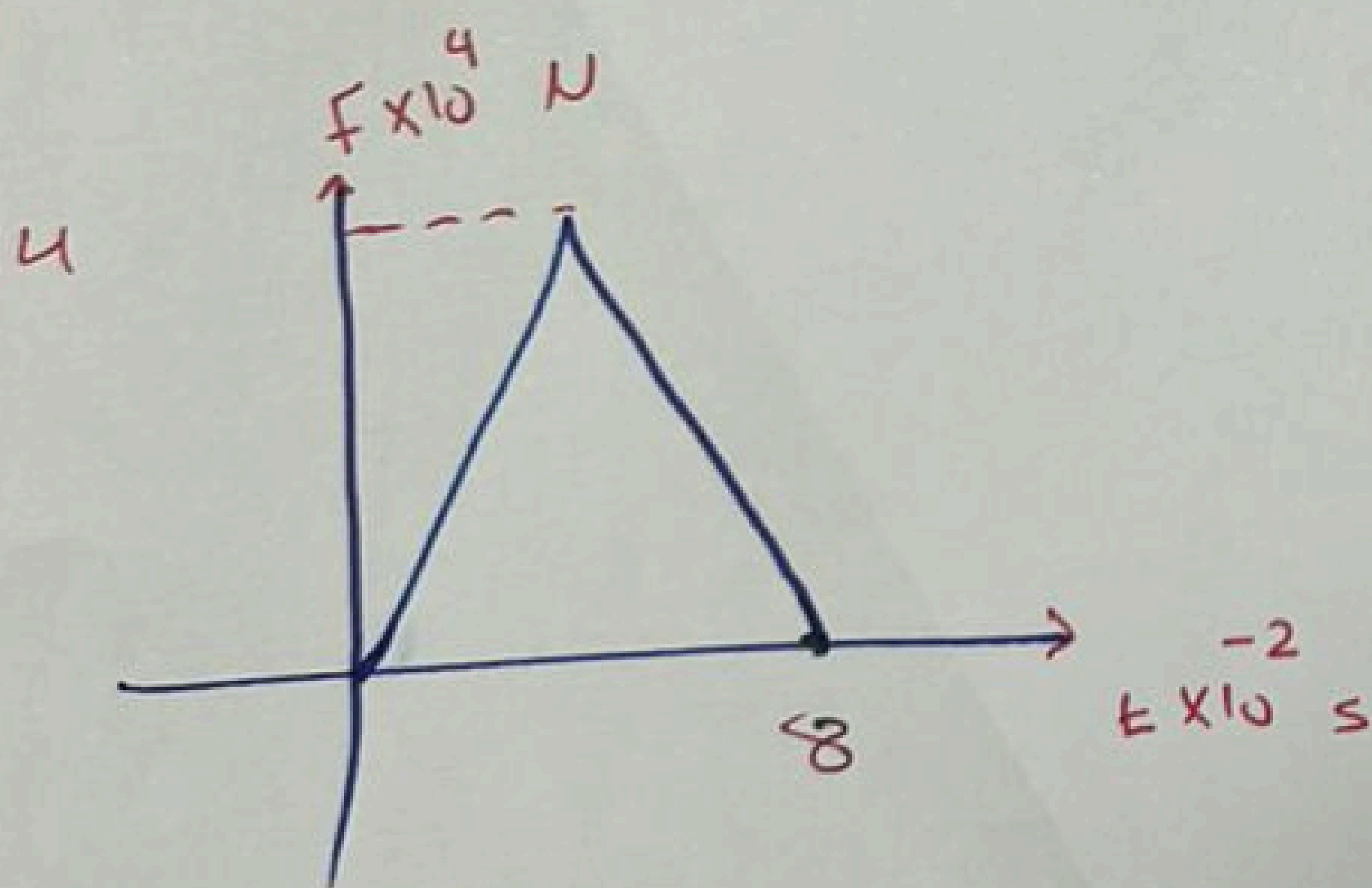
$$Q_1 = 2Q_2$$

$$40 \text{ nC} = 2Q_2$$

$$Q_2 = 20 \text{ nC}$$

$$Q_3 = 20 \text{ nC}$$

$$Q_2 = Q_3$$



$$m = 5 \times 10^{-2} \text{ kg}$$

$$\Delta P = \frac{\Delta P}{\Delta t} = \frac{\frac{1}{2} (8 \times 10^{-2}) (4 \times 10^4)}{8 \times 10^{-2}} = 2 \times 10^4 \text{ N}$$

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$$v_i = 0$$

$$\Delta P = m(v_f - v_i)$$

$$\frac{1}{2} (8 \times 10^{-2}) (4 \times 10^4) = 5 \times 10^{-2} (v_f - 0)$$

$$16 \times 10^2 = 5 \times 10^{-2} (v_f)$$

$$v_f = 3.2 \times 10^2 \text{ m/s}$$

4

$$m_A = m$$

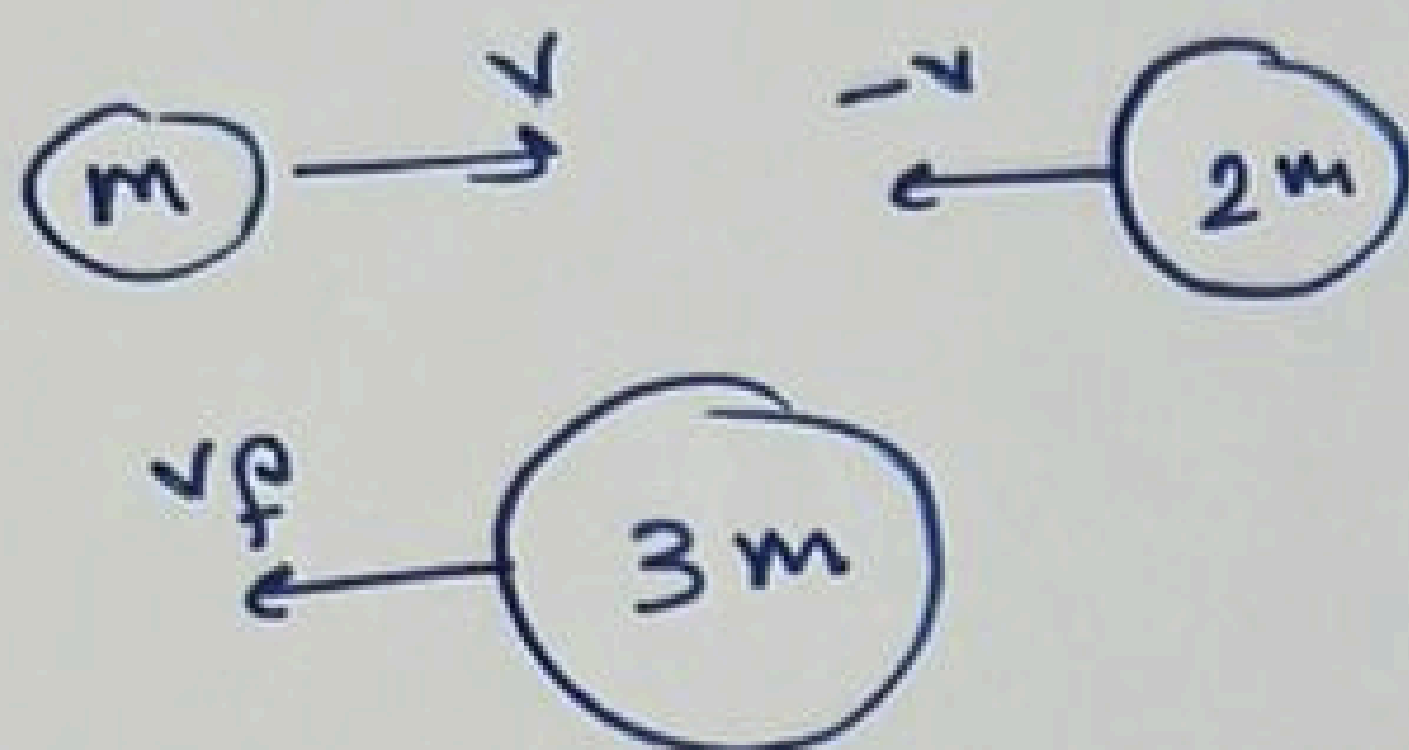
$$m_B = 2m$$

$$v_{Ai} = v \text{ و } X^+$$

$$v_{Bi} = -v$$

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الكم (الضاد) بعد التصادم



$$\sum P_i = \sum P_f$$

$$m(v) + 2m(-v) = (3m) v_p$$

$$-mv = 3m v_p$$

$$v_p = \frac{-v}{3}$$

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$$\sum KE_i = KE_{iA} + KE_{iB}$$

الطاقة
الحركية
للنظام قبل
التصادم

$$= \frac{1}{2} m_A v_{Ai}^2 + \frac{1}{2} m_B v_{Bi}^2$$

$$= \frac{1}{2} m v^2 + \frac{1}{2} 2m (-v)^2$$

$$= \frac{1}{2} m v^2 + \frac{m v^2}{1 \times 2}$$

$$= \frac{m v^2}{2} + \frac{2m v^2}{2} = \frac{3m v^2}{2}$$

$$d' = 2d$$

الطاقة
الحركية
للنظام بعد
التصادم

$$Q = \text{const}$$

$$C' = \frac{Q \cdot A}{d'} \rightarrow \frac{C}{2}$$

$$Q \rightarrow \text{const}$$

$$V' = \frac{Q}{C'} \rightarrow 2V$$

$$PE = \frac{1}{2} \frac{Q^2}{C'} \rightarrow 2PE$$

$$E = \frac{\Delta V'}{d'} = \frac{2V}{\frac{d}{2}} = \frac{4V}{d}$$

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الطاقة
الحركية

إذاً الجهد هو موجود
بالتالي ان نختار
PE